

Will start at 9:31 am :)

ECON1050 Tutorial 1

Sets, Logic & Algebra Review

Semester 2 2026

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3 & 4 August

Today's Plan

- ▶ Intro
- ▶ Recap: core concepts from Lecture 1
- ▶ Tutorial questions: Q1–Q5

A bit about Xavier Xi Zhang

- ▶ My name: Xavier Xi Zhang
 - ▶ You can call me **Xavier**, **Xav**, **Xi**, or **X**.
- ▶ Economics Honours student (ECON/MATH student before that).
- ▶ Personal website: xavierxizhang.github.io
 - ▶ Tutorial slides will be uploaded under **Teaching**.
- ▶ Questions are welcome (ECON, MATH, or beyond).
- ▶ My consultation hours are every Monday from 11 am – 12:30 pm, at 39-125B.

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ECON2050 is hardest econ course

- ▶ This is one of the TWO maths courses offered specifically for ECON students.
- ▶ Three parts:
 - ▶ Foundations: sets, logic, algebra, solving equations, functions...
 - ▶ Calculus
 - ▶ Matrix algebra
- ▶ Some tips:
 - ▶ As long as you know how to solve all the tutorial questions, you should be able to get a 7.
 - ▶ So make sure you take the tutorials seriously!
- ▶ If you enjoy this course, I'd recommend taking more maths courses.

Recap: Sets and notation

Sets and elements

A **set** is a collection of elements. Membership is written as

$$a \in A \quad \text{or} \quad a \notin A.$$

A set may be listed using braces or described by a defining property.

$$B = \{1, 2, 3, 4\}$$

$$A = \{1, 2, 3\}$$

$$\underline{A \subset B}$$

Subsets

$$A \subset B$$

means that every element of A belongs to B , and B contains at least one additional element. The empty set is denoted by $\emptyset$.

Set-builder notation

$$\{x : x \text{ satisfies a stated property}\}.$$

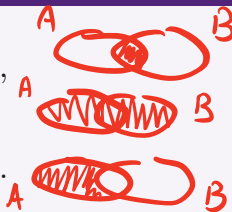
Recap: Set operations

Intersection, union and difference

$$A \cap B = \{x : x \in A \text{ and } x \in B\},$$

$$A \cup B = \{x : x \in A \text{ or } x \in B\},$$

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

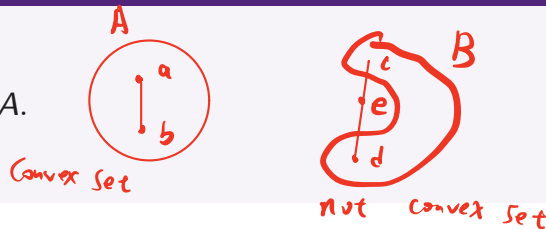


Universal set and complement

If Ω is the universal set, the complement of A is

$$A' = \Omega \setminus A.$$

Two sets are **disjoint** when $A \cap B = \emptyset$.



Convex set

A set is convex when the line segment joining any two points in the set lies entirely inside the set.

Recap: Propositions and logical relations

Proposition

A proposition is a statement that is either true or false.

ex.

I'm Xavier ✓

I'm Messi ✗

Implication and converse

means "if P , then Q ." Its converse is

$$\underline{P} \Rightarrow \underline{Q}$$

$$Q \Rightarrow P.$$

P condition to derive Q

$$P \Rightarrow Q$$

$$Q \Rightarrow P$$

The truth of an implication does not automatically imply the truth of its converse.

Logical equivalence

means that both $P \Rightarrow Q$ and $Q \Rightarrow P$ are true.

$$P \Leftrightarrow Q$$

$$P \Rightarrow Q \text{ and } Q \Rightarrow P$$

$$P \Leftrightarrow Q$$

ex. $P: X-2=0$ $Q: X=2$

Recap: Necessary and sufficient conditions

When $P \Rightarrow Q$

- ▶ P is a **sufficient condition** for Q .
- ▶ Q is a **necessary condition** for P .
- ▶ Equivalent wording includes “ P only if Q ” and “ Q if P .”

P $\Rightarrow$ Q

When $P \Leftrightarrow Q$

P is a necessary and sufficient condition for Q , and vice versa. This is read as “ P if and only if Q ” or “ P iff Q .”

Recap: Mathematical proofs

Theorems and lemmas

Mathematical results are commonly stated as implications

$$P \Rightarrow Q,$$

where P contains the premises and Q is the conclusion.

$P \Rightarrow Q$
ex. we know $\frac{P}{a > b}$
prove $\frac{a - b > 0}{Q}$
 $P \Rightarrow Q$

Proof methods

- ▶ **Direct proof:** derive the conclusion through a sequence of valid logical steps.
- ▶ **Proof by contradiction:** assume the required conclusion is false and derive a contradiction.
- ▶ **Mathematical induction:** verify a base case, assume the result for $n = k$, and prove it for $n = k + 1$.

Recap: Algebraic rules

Basic laws

$$\begin{aligned}a + b &= b + a, & ab &= ba, \\(a + b) + c &= a + (b + c), & a(bc) &= (ab)c, \\a(b + c) &= ab + ac.\end{aligned}$$

Powers

For appropriate real numbers and integer exponents,

$$a^r a^s = a^{r+s}, \quad \frac{a^r}{a^s} = a^{r-s}, \quad (ab)^r = a^r b^r, \quad a^{-r} = \frac{1}{a^r}.$$

Quadratic identities

$$(a + b)^2 = a^2 + 2ab + b^2, \quad (a - b)^2 = a^2 - 2ab + b^2, \quad (a - b)(a + b) = a^2 - b^2.$$

Recap: Fractions and fractional powers

Fractions

Denominators must be non-zero. The standard rules are

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a/b}{c/d} = \frac{ad}{bc}.$$

Fractional powers

$$a^{p/q} = \left(a^{1/q}\right)^p.$$

The symbol $\sqrt{a} = a^{1/2}$ denotes the **principal square root**, which is non-negative.

Recap: Inequalities

Strict and weak inequalities

$$a > b, \quad a < b, \quad a \geq b, \quad a \leq b.$$

The statement $a > b$ is equivalent to $a - b > 0$.

Manipulation rules

- ▶ Adding or subtracting the same number preserves the direction.
- ▶ Multiplying or dividing by a positive number preserves the direction.
- ▶ Multiplying or dividing by a negative number reverses the direction.

$$\begin{array}{l} a > b \\ - \\ a \times (-1) < b \times (-1) \end{array}$$

Recap: Sign diagrams and intervals

Sign diagrams

To solve a product or rational inequality:

1. move all terms to one side and factor where possible;
2. identify zeros and points where the expression is undefined;
3. determine the sign on each interval;
4. include or exclude boundary points according to the inequality.

Intervals

(a, b) , $[a, b]$, $(a, b]$, $[a, b)$.

Round brackets exclude an endpoint; square brackets include it.

ex. $X^2 < 1$

1. $X^2 - 1 < 0$

2. $X^2 - 1 = (X+1)(X-1) = 0 \Rightarrow X = -1, 1$

Sign diagram Method

3. Sign diagram

X	$(-\infty, -1)$	-1	$(-1, 1)$	1	$(1, \infty)$
$X-1$	-	-	-	0	+
$X+1$	-	0	+	+	+
$(X+1)(X-1)$	+	0	-	0	+

4. $X \in (-1, 1)$

Recap: Absolute values

Definition

For any real number a ,

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

The absolute value is the distance from the origin on the real number line.

Absolute-value inequalities

For $b > 0$,

$$|x| < b \Leftrightarrow -b < x < b,$$

$$|x| > b \Leftrightarrow x < -b \text{ or } x > b.$$

Tutorial Q1

Question 1

Use a sign diagram to determine for which values of x each inequality holds.

(a)

$$(x - 1)^3(x + 2) \leq 0.$$

5 mls

(b)

$$2 < \frac{3x + 1}{2x + 4}.$$

$$(a) \quad (x-1)^3(x+2) \leq 0$$

$$1. \quad (x-1)^3(x+2) \leq 0$$

$$2. \quad (x-1)^3(x+2) = 0 \Rightarrow x = -2, 1$$

3. Sign diagram:

x	$(-\infty, -2)$	-2	$(-2, 1)$	1	$(1, +\infty)$
$(x-1)^3$	-	-	-	0	+
$x+2$	-	0	+	+	+
$(x-1)^3(x+2)$	+	0	-	0	+

$$4. \quad x \in [-2, 1]$$

$$(b) \quad 2 < \frac{3x+1}{2x+4}$$

$$1. \quad 2 - \frac{3x+1}{2x+4} < 0$$

$$2. \quad 2 - \frac{3x+1}{2x+4} = \frac{4x+8-3x-1}{2x+4} = \frac{x+7}{2x+4} < 0$$

$$\Rightarrow (x+7)(2x+4) < 0$$

$$(x+7)(2x+4) = 0 \Rightarrow x = -7, -2$$

3. Sign diagram:

x	$(-\infty, -7)$	-7	$(-7, -2)$	-2	$(-2, \infty)$
$x+7$	-	0	+	+	+
$2x+4$	-	-	-	0	+
$(x+7)(2x+4)$	+	0	-	0	+

$$4. \quad x \in (-7, -2)$$

Tutorial Q1(a): Sign diagram

Knowledge used

- ▶ A factor raised to an odd power has the same sign as the factor.
- ▶ Use zeros to divide the real line into sign intervals.
- ▶ Include zeros when the inequality is weak.

Solution

The critical values are $x = -2$ and $x = 1$. Since $(x - 1)^3$ has the same sign as $x - 1$,

x	$(-\infty, -2)$	-2	$(-2, 1)$	1	$(1, \infty)$
$(x - 1)^3$	$-$	$-$	$-$	0	$+$
$x + 2$	$-$	0	$+$	$+$	$+$
$(x - 1)^3(x + 2)$	$+$	0	$-$	0	$+$

Therefore,

$$x \in [-2, 1].$$

Tutorial Q1(b): Knowledge used

Knowledge used

- ▶ Rearrange the inequality into one rational expression.
- ▶ Use numerator zeros and denominator zeros to divide the real line into sign intervals.
- ▶ A denominator zero is excluded from the solution.
- ▶ Do not multiply by an expression of unknown sign without checking whether the inequality reverses.

Tutorial Q1(b): Solution

Solution

$$2 < \frac{3x+1}{2x+4} \Leftrightarrow \frac{3x+1}{2x+4} - 2 > 0 \Leftrightarrow \frac{-x-7}{2x+4} > 0.$$

With critical values -7 and -2 ,

x	$(-\infty, -7)$	-7	$(-7, -2)$	-2	$(-2, \infty)$
$-x-7$	+	0	-	-	-
$2x+4$	-	-	-	0	+
$\frac{-x-7}{2x+4}$	-	0	+	undefined	-

Thus,

$$-7 < x < -2 \quad \text{or} \quad x \in (-7, -2).$$

Tutorial Q2

Question 2

Show that if

$$a + \overset{P_1}{b} > 0,$$

3 mins

then

$$\overset{P_2}{a} > b \implies \overset{Q}{a^2} > b^2.$$

$$P_1, P_2 \Rightarrow Q$$

$$\text{Proof: } P_2 : a > b \Rightarrow a - b > 0$$

$$P_1 \times P_2 : (a+b) \cdot (a-b) > 0$$

$$a^2 + ab - ab - b^2 > 0$$

$$a^2 - b^2 > 0$$

$$a^2 > b^2 \Leftrightarrow Q$$



Tutorial Q2: Direct proof

Knowledge used

- ▶ Direct proof: begin with the assumptions and derive the conclusion.
- ▶ $a > b$ is equivalent to $a - b > 0$.
- ▶ The product of two positive numbers is positive.
- ▶ Difference of squares: $(a + b)(a - b) = a^2 - b^2$.

Solution

From $a > b$,

$$a - b > 0.$$

Together with $a + b > 0$, this gives

$$(a + b)(a - b) > 0 \Leftrightarrow a^2 - b^2 > 0 \Leftrightarrow a^2 > b^2.$$

Question 3

For each implication, decide:

1. whether the implication is true;
2. whether the converse implication is true.

$$P \Rightarrow Q ?$$

$$Q \Rightarrow P ?$$

(a)

P

Q

$$x = 5 \text{ and } y = -3 \Rightarrow x + y = 2.$$

(b)

$$(x - 3)^2(y + 2) > 0 \Rightarrow y > -2.$$

$$(a) \quad P: x=5 \text{ and } y=-3, \quad Q: x+y=2$$

$$P \Rightarrow Q? : x=5, y=-3, \quad \underline{x+y=5-3=2} \quad \checkmark$$

$$Q \Rightarrow P? : x+y=2 \quad \left\{ \begin{array}{l} x=0, y=2 \\ x=1, y=1 \end{array} \right. \quad \nRightarrow x=5, y=-3 \quad \times$$

$$(b) P: (x-3)^2(y+2) > 0, Q: y > -2$$

$$P \Rightarrow Q? : \frac{(x-3)^2(y+2) > 0}{>0} \Rightarrow \frac{y+2 > 0}{y > -2} \quad \checkmark$$

$$\underline{Q}$$

$$Q \Rightarrow P? : y > -2 \Rightarrow y+2 > 0, X=3 \Rightarrow X-3=0 \Rightarrow (X-3)^2=0$$

$$(x-3)^2 \cdot (y+2) = 0 \cdot (y+2) = \underline{0}$$

$$P: (x-3)^2(y+2) > 0 \quad \times$$

Big set $\nRightarrow$ Small set

Small set $\Rightarrow$ Big set

Tutorial Q3(a): Implication and converse

Knowledge used

- ▶ To verify an implication, apply its premises and check the conclusion.
- ▶ The converse of $P \Rightarrow Q$ is $Q \Rightarrow P$.
- ▶ One counterexample is sufficient to show that an implication is false.

Solution

The implication is true because

$$x + y = 5 + (-3) = 2.$$

Its converse is

$$x + y = 2 \Rightarrow x = 5 \text{ and } y = -3.$$

The converse is false: $x = 7$ and $y = -5$ give $x + y = 2$ but do not satisfy the original premises. Hence,

the implication is true, but its converse is false.

Tutorial Q3(b): Implication and converse

Knowledge used

- ▶ A square is always non-negative; strict positivity also requires the squared factor to be non-zero.
- ▶ Test the converse separately and use a counterexample when it is false.

Solution

Because $(x - 3)^2 \geq 0$ for every real x ,

$$(x - 3)^2(y + 2) > 0 \Leftrightarrow x \neq 3 \text{ and } y > -2.$$

Hence the original implication is true. Its converse is false because $x = 3$ makes the product equal to zero even when $y > -2$.

Original implication: true. Converse implication: false.

Tutorial Q4

Question 4

$\Leftrightarrow$ $\Leftarrow$ $\Rightarrow$

Fill each blank with “iff”, “if”, or “only if”, and use the corresponding implication sign.

(a)

$x = \sqrt{4}$ $\boxed{\text{iff} \Leftrightarrow}$ $x = 2$.

$P \Rightarrow Q?$
 $P \Leftarrow Q?$

(b)

$x(x+3) < 0$ $\boxed{\text{only if} \Rightarrow}$ $x > -3$.

(c)

$x^2 < 9$ $\boxed{\text{only if} \Rightarrow}$ $x < 3$.

(a) $P: X = \sqrt{4} = 2$ $Q: X = 2$
 $P \Rightarrow Q \checkmark$, $Q \Rightarrow P \checkmark$ $P \Leftrightarrow Q$

(b) $P: x(x+3) < 0$ $Q: x > -3$

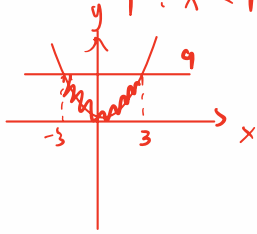


$P: x \in (-3, 0)$ $Q: x \in (-3, \infty)$
 Smaller set bigger set
 $(-3, 0) \subset (-3, \infty)$

$P \Rightarrow Q \checkmark$, $Q \not\Rightarrow P$

(c) $P: x^2 < 9$

$Q: x < 3$



$P: x \in (-3, 3)$, $Q: x \in (-\infty, 3)$
 Smaller set bigger set
 $(-3, 3) \subset (-\infty, 3)$

$P \Rightarrow Q \checkmark$, $Q \not\Rightarrow P$

Tutorial Q4: Knowledge used

Knowledge used

- ▶ “ P only if Q ” means $P \Rightarrow Q$.
- ▶ “ P if Q ” means $Q \Rightarrow P$.
- ▶ “ P iff Q ” means $P \Leftrightarrow Q$.
- ▶ $\sqrt{a}$ denotes the non-negative principal square root.
- ▶ Sign diagrams and interval notation identify the exact solution set of an inequality.

Tutorial Q4: Answers

Solution

(a) By the principal-square-root convention,

$$x = \sqrt{4} \Leftrightarrow x = 2.$$

Therefore the correct word is *iff*.

(b) Since

$$x(x+3) < 0 \Leftrightarrow -3 < x < 0,$$

we have $x(x+3) < 0 \Rightarrow x > -3$, but not conversely. Therefore,

$$x(x+3) < 0 \text{ only if } x > -3.$$

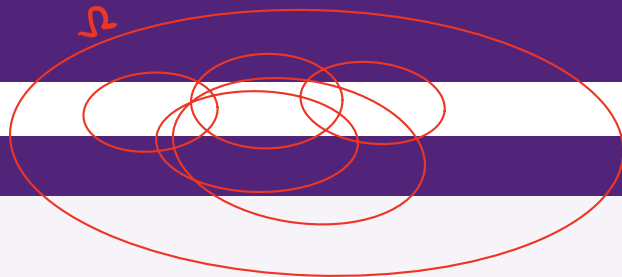
(c) Since

$$x^2 < 9 \Leftrightarrow -3 < x < 3,$$

we have $x^2 < 9 \Rightarrow x < 3$, but not conversely. Therefore,

$$x^2 < 9 \text{ only if } x < 3.$$

Tutorial Q5



Question 5

Let Ω be the set of all students at a university. Let

F = female students, M = mathematics students, C = choir members,

B = biology students, T = tennis players.

Write the following statements in set terminology.

Students who study both B and M

$$B \cap M$$



(a) All biology students who are mathematics students.

(b) There are female biology students in the choir.

Students

$$\begin{matrix} F \\ B \\ C \end{matrix}$$

$$F \cap B \cap C \neq \emptyset$$

(c) No tennis player studies biology.

Students

$$\begin{matrix} T \\ B \end{matrix}$$

$$T \cap B = \emptyset$$

(d) Students who neither play tennis nor belong to the choir all study biology.

$$T \cup C$$



$$\Omega \setminus (T \cup C) \subset B$$

Tutorial Q5: Knowledge used

Knowledge used

- ▶ “and” corresponds to intersection, while “or” corresponds to union.
- ▶ “neither A nor B ” corresponds to the complement of $A \cup B$.
- ▶ “there are” means the relevant set is non-empty.
- ▶ “no members” means the relevant intersection is empty.
- ▶ “all members of one set belong to another” is expressed using a subset relation.

Tutorial Q5: Answers

Solution

- (a) $B \cap M$.
- (b) $F \cap B \cap C \neq \emptyset$.
- (c) $T \cap B = \emptyset$.
- (d) $\Omega \setminus (T \cup C) \subset B$.